INCH-POUND

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SUPERSEDING
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PERFORMANCE SPECIFICATION

PLASTIC RODS AND TUBES, POLYETHYLENE

This specification is approved for use by all departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers low, medium, and high density polyethylene rods and tubes for general purpose, dielectric, and weather resistant applications.

1.1.1 Coverage. This specification covers polyethylene rods and tubes made from material conforming to L-P-390.

1.2 Classification. Polyethylene rods and tubes will be of the applicable type, class, and grade (see 3.1 and 6.2) in accordance with table I, as follows..

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Industrial Hardware Division (ATTN: Code FHTE), 700 Robbins Avenue, Philadelphia, PA 19111-5096 or email trpsptspecspa@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .

AMSC/NA FSC 9330

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Table I. Types, Classes and Grades of Rods and Tubes

type	Application	Class 1/	Grade	Color
I	General purpose	L and M	1 2	Natural and colors, including black
		H	1 2 3 4 5	
II	Dielectric	L	1 2 3 4	Natural Colors, including black Natural Colors, including black
		M	1 2	Natural Colors, including black
		H	1 2 3 4 5	Natural and colors, including black
III	Weather resistant	L	1 2 3 4	Black (carbon .5 ±.1 percent by weight Black (carbon 2.5 ±.5 percent by weight
		M	1 2 3	Black (carbon .5 ±.1 percent by weight Black (carbon 2.5 ±.5 percent by weight
		H	1 2	

1/ L = low density; M = medium density; and H = high density.

Form A - Rods
Form B - Tubs

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-390 - Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium and High Density)

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STANDARDS

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of these documents are available online at http://quicksearch.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues in effect on the date of invitation for bids or request for proposal will apply.

American Society for Quality (ASQ):

ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes..

(Copies of this document are available online from http://asq.org or from the ASQ Distribution Center, 5131 S. Third Street, Milwaukee, Wisconsin 53207-6028.)

ASTM International Standards:

ASTM D618	- Standard Practice for Conditioning Plastics and Electrical Insulating Materials for Testing
ASTM D1238	- Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer
ASTM D1505	- Standard Test Method for Density of Plastics By the Density-Gradient Technique

(Copies of these documents are available from http://www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 Materials.. Unless otherwise specified by the procuring activity (see 6.2), type I material of L-P-390 shall be used for extrusion of rods and tube

3.2 Property values. The rods and tubes shall conform to the property values for density and melt index specified L-P-390 (see 4.3.2).

3.3 Dimensions and tolerances.

3.3.1 Diameter and wall thickness. The diameter of rods shall be the nominal diameter specified by the procuring activity (see 6.2) and tolerances shall be as specified in table II. The outside diameter and wall thickness of tubes shall be as specified by the procuring activity (see 6.2) and tolerances shall be as specified in table III.

Table II, Rod diameter tolerances, including out of roundness

1/	Tolerance		1/	Tolerance	
Nominal diameter, inches	Plus	Minus	Nominal diameter, inches	Plus	Minus
.125 (1/8)	.010	.010	1.250 (1-1/4)	.025	.020
.187 (3/16)	.010	.010	1.500 (1-1/2)	.025	.020
.250 (1/4)	.010	.010	2.000	.040	.020
.375 (3/8)	.010	.018	2.500 (2-1/2)	.048	.020
.500 (1/2)	.010	.018	3.000	.250	.000
.625 (5/8)	.010	.018	4.000	.250	.000
.750 (3/4)	.010	.018	5.000	.250	.000
.875 (7/8)	.010	.020	6.000	.250	.000
1.000	.010	.020	7.000	.250	.000

1/ Intermediate diameters shall conform to the tolerances of the larger diameters.

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Table III. Tolerances for tube diameter and wall thickness.

Nominal outside diameter of tubes, inches 1/	Tolerances , plus or minus, inches 2/					
	On outside diameter, inches	On outside On Wall thickness, inches 3/				
		.031 (1/32)	.062 (1/16)	.093 (3/32)	.125 (1/8)	.187 (3/16)
.375 (3/8) 4/	.0075	.005	.006	.008	.009	-----
.500 (1/2)	.010	.005	.007	.009	.010	.012
.625 (5/8)	.011	.005	.007	.009	.010	.012
.750 (3/4)	.0125	.005	.007	.009	.010	.012
1.000	.015	.006	.008	.010	.010	.013
1.250 (1-1/4)	.0175	.006	.008	.010	.010	.013
1.500 (1-1/2)	.020	.006	.008	.010	.010	.014
2.000	.025	.006	.009	.010	.011	.015
2.750 (2-3/4)	.035	.006	.010	.010	.012	.015
2.500 (2-1/2)	.030	.006	.009	.010	.012	.015

1/ Intermediate sizes shall conform to the tolerances of the larger size.

2/ Tolerances for diameter apply to the average outside diameter as determined by averaging two mutually perpendicular diameters in the same cross sectional plane. Tolerance for wall thickness applies to the maximum and minimum wall thickness measured at any point around the circumference and includes eccentricity.

3/ Tolerances for wall thickness greater than .187 (3/16) inch shall be as specified by the procuring activity.

4/ Tolerances for tubes smaller than .375 (3/8) inch or larger than 2.750 (2-3/4) inches outside diameter, shall be as specified by the procuring activity (see 6.2).

3.3.2 Length of rods. Unless otherwise specified by the procuring activity (see 6.2), rods shall be furnished in lengths of 48 or 72 inches depending upon which is more nearly standard for the diameter specified. When both lengths are standard, the longer one shall be preferred. Tolerances for length shall be plus 1.0, minus 0.0 inches for rods 1 inch or less in diameter, and plus 2.0, minus 0.0 inches for rods greater than 1 inch in diameter (see 6.2). Rods shall be supplied in straight lengths.

3.3.3 Length of tubes. Unless otherwise specified by the procuring activity (see 6.2), tubes shall be furnished in straight lengths of 72 plus 1.0, minus 0.0 inches (see 6.2).

3.4 Color. The color of the rods and tubes shall be as specified by the procuring activity (see 6.2) .

3.5 Centerless grinding. When specified by the procuring activity , rods and tubes shall be extruded over size and then centerless ground to the outside diameter specified. The rods or tubes shall be held to the outside diameter tolerance and surface finish specified in the contract or purchase order. The wall thickness shall be as specified, plus or minus the acceptable tolerances.

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3.6 Certification. The manufacturer shall present certification that the material used in preparing the rods and tubes conforms to the applicable type, class, and grade of material specified in L-P-390 (see 6.2). The certificate shall state the material designation, color, and type, grade, and class of L-P-390 to which it conforms. The certificate shall be presented prior to, or at the time of delivery of the lot, and shall be signed by a responsible agent of the supplier, and shall be accompanied by evidence of the agent's authority to bind his principal.

3.7 Workmanship. The rods and tubes shall be free from kinks (see 6.3), blisters, cracks, bubbles, discolorations, craze, surface scratches that form definite indentations, wrinkles, dents, die or heat marks.

4. VERIFICATION

4.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced, subsidiary specifications and standards to the extent specified, In the event of conflict, this specifications hall govern.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in ASQ Z1.4, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all rods of one nominal diameter, or tubes of one nominal outside diameter and wall thickness of the same type, class and grade of material, submitted for delivery at one time.

4.3 Inspection of the rods and tubes.

4.3.1 Examination of the rods and tubes. Examination shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purposes of determining the sample size in accordance with ASQ Z1.4, shall be expressed in units of packages of plastic rods or tubes as applicable, for examination in 4.3.1.1, 4.3.1.2, 4.3.1.3 and-in units of shipping containers for examination under 4.3.1.4.

4.3.1.1 Examination of the rods and tubes for defects in color, centerless grinding, and workmanship. The sample unit for this examination, specified in table IV, shall be one rod or one tube, as applicable. Not more than five sample units shall be taken from any one package of rods or tubes.

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Table IV. Examination for defects in color, centerless grinding, and workmanship

Examine	Defect
Color and Centerless grinding	Color improper Centerless grinding improper
Workmanship	Not free from kinks, blisters, cracks, bubbles, discolorations, craze, surface scratches forming indentations, wrinkles, dents, die or heat marks

4.3.1.2 Examination of the rods and tubes for dimensional defects. The sample unit for this examination, specified in table V, shall be one rod or tube, as applicable

Table V. Examination for dimensional defects

Examine	Defect
Diameter of rods	Nominal diameter improper or varies more than tolerances specified in table II
Outside diameter and wall thickness of tubes-	Improper outside diameter or wall thickness. Tolerances not within that specified in table III.
Length of rods	Improper length. Not within tolerances specified for length of rods.
Length of tubes	Improper length. Not within tolerances specified for length of tubes.

4,3,1.3 Examination of the rods and tubes for defects in the count for package. The sample unit for this examination, specified in table VI shall be one package.

Table VI. Examination for defects in the count for package

Examine	Defect
Rods and tubes	Average count per package less than specified

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4.3.1.4 Examination of preparation for delivery requirements. An examination shall be made i n accordance with table VII to determine that packaging, packing and marking comply with the requirements of Section 5. The sample unit for this examination shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Table VII. Examination of preparation for delivery

Examine	Defect
Packaging	Not level specified; not in accordance with contract requirements.
	Rods or tubes not unit wrapped and packaged as specified.
	Packaging material not as specified; closures not accomplished by specified or required methods or materials.
Packing	Not level specified; not in accordance with contract requirements.
	Any nonconforming component; component missing, damaged or otherwise defective affecting serviceability.
	Container not as specified; closures not: accomplished by specified or required methods of materials.
	Inadequate application of components, such as: incomplete closures of case liners or container flaps, loose or inadequate strappings, bulged or distorted containers, improper taping or inadequate stapling
Count	Less than specified or indicated quantify of packages per shipping container.
Weight	Gross or net weight exceeds specified requirements.
Marking	Interior or exterior markings (as applicable) omitted, illegible, incorrect, incomplete, of improper size, location, sequence, method of application, or not in accordance with contract' requirements

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4.3.1.5 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptable quality levels (AQLs) expressed in defects per 100 units, shall be as follows:

Examination Paragraph	Inspection Level	AQL
4.3.1.1	I	1.5
4.3.1.2	S-3	2.5
4.3.1.3	S-2	---
4.3.1.4	S-2	4.0

4.3.2 Testing. The rods or tubes shall be tested for the density and melt index characteristics specified in 3.2 for each lot submitted for inspection, in accordance with the test methods specified herein. The lot size for the purpose of determining sample size for testing shall be expressed in units of packages of rods or tubes. The sample unit shall consist of approximately two feet of rod or tube. The inspection level shall be S-1 with an acceptance number of 0. The results for each test shall be the averaged results of the specimens.

4.4 Test methods.

4.4.1 Specimen preparation. Specimens shall be die cut or machined from rods or tubes for density determinations. For melt index determinations, specimens may be in any of the forms specified in ASTM D 1238.

4.4.2 Conditioning test specimens. The specimens shall be conditioned i n accordance with procedure A of ASTM method D 618. Testing shall be at 23º ± 2ºC (73.4º ± 3.6ºF) and 50 ± 5 percent relative humidity.

4.4.3 Density. Duplicate determinations shall be made using two separate specimens. Testing shall be in accordance with ASTM D 1505.

4.4.4 Melt index. Three determinations shall be made for melt index (flow rate) in accordance with ASTM method D1238, using condition E.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The bulk of polyethylene rods and straight length tubes are intended for use as machine shop stock for machining into components of electrical and corrosion resistant equipment and items. Small diameter tubes, usually furnished in coils, may be used as laboratory tubing for carrying water and corrosive liquids. Smaller diameter rods may be used for decorative beading.

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6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Type if other than type I; class, and grade of material required (see 1.2).
c. Outside diameter required (see 3.3.1).
d. Tube wall thickness required (see 3.3.1).
e. Length required (see 3.3.2 and 3.3.3).
f. Color required (see 3.4).
g. Centerless grinding an d surface finish, if required (see 3.5).
h. Tube diameter and wall thickness tolerances, when centerless grinding is specified (see 3.5).
i. Packaging and packing required including unit packaging quantities, if required (see 5.1 and 5.2).

6.2.2 Contract data requirements. Data specified in paragraphs 3.2 and 3.6 will be listed directly on a DD Form 1423 incorporated into the contract.

6.3 Kinked and bowed. Rods and tubes containing a bend that is not a uniform bow are considered kinked. Rods and tubes uniformly curved over the entire length are considered bowed. Unless otherwise specified, bowed rods and tubes are acceptable, and kinked rods and tubes are unacceptable. Numerical limits for bowing may be specified by the procuring activity.

6.4 Degradation of material. Extrusion under excessive temperature conditions may cause degradation of the material as evidenced by a change in appearance and an increase in melt index.

6.5 Coils. The procuring activity may specify that small diameter tubes by supplied in coils of specified lengths. Usual coil lengths are 50 feet and 100 feet. Because of the tendency for coiled tubes to contain coil set and kinking, it is recommended that only tubes supplied in straight lengths be ordered for use as machine shop stock.

6.6 Subject term (key word) listing:

Low polyethylene
Medium polyethylene
High density polyethylene
Dielectric
Weather resistant

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MR
Navy - AS
Air Force –20

Preparing Activity:
DLA - IS

(Project 9330-2014-003)

Review Activity:
Army – AR, EA, GL, MI

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